Water Pollution

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What is Water Quality?

- Water quality refers to the physical, chemical, and bacteriological properties of water, which are measured based on specific usage requirements and quality concerns. It is determined by a set of 16 basic parameters that are essential for ensuring the quality of water for various purposes such as domestic, agriculture, industrial, and fisheries.

Source: 1. Water Quality, Sciencedirect, https://www.sciencedirect.com/topics/earth-and-planetary-sciences/water-quality
2. Image: Tektelic, https://tektelic.com/expertise/smart-water-quality-monitoring-why-is-it-better-than-traditional-methods/

Distribution of Water

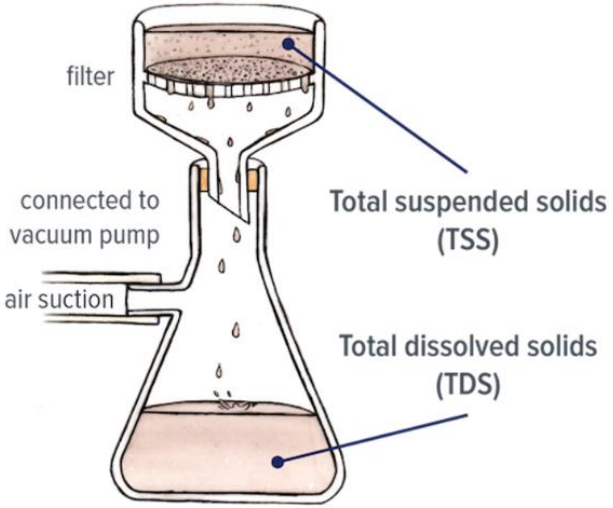
- Over 97% of the total water supply is contained in the oceans and the other saline bodies of water and isn't readily usable for most purposes.
- For human livelihood and to support their technical and agricultural requirements humans depend on 0.62% remaining water found in freshwater lakes, rives and ground waters.

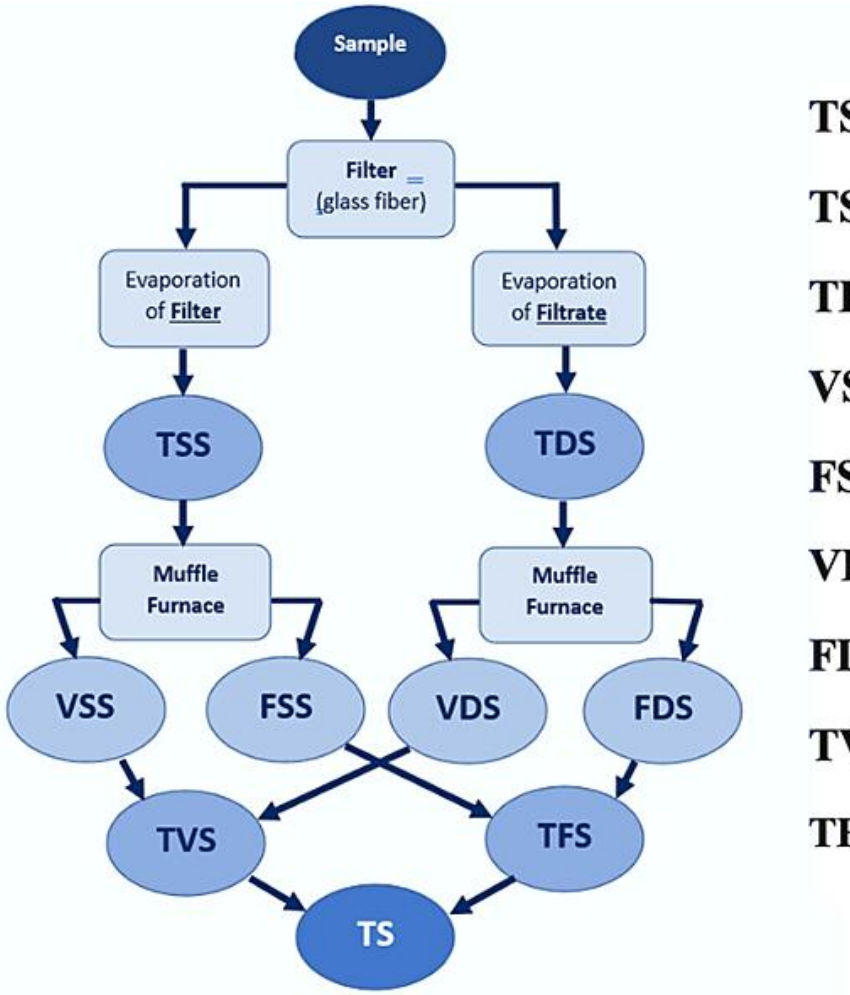
Location	Volume, 10^{12} m^3	% of total
Land areas		
Freshwater lakes	125	0.009
Saline lakes and inland seas	104	0.008
Rivers (average instantaneous volume)	1.25	0.0001
Soil Moisture	67	0.005
Groundwater (above depth of 4000 m)	8, 350	0.61
Ice caps and glaciers	29, 200	2.14
Total land area (rounded)	37, 800	2.8
Atmosphere (Water Vapor)	13	0.001
Oceans	1,320,000	97.3
Total all locations (Rounded)	**1,360,000**	**100**

Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Total Solids

- Dissolved solids consist of calcium, chlorides, nitrate, phosphorus, iron, sulfur, and other ions particles that will pass through a filter with pores of around 2 microns (0.002 cm) in size. Suspended solids include silt and clay particles, plankton, algae, fine organic debris, and other particulate matter. These are particles that will not pass through a 2-micron filter.





TS = total solids

TSS = total suspended solids

TDS = total dissolved solids

VSS = volatile suspended solids

FSS = fixed suspended solids

VDS = volatile dissolved solids

FDS = fixed dissolved solids

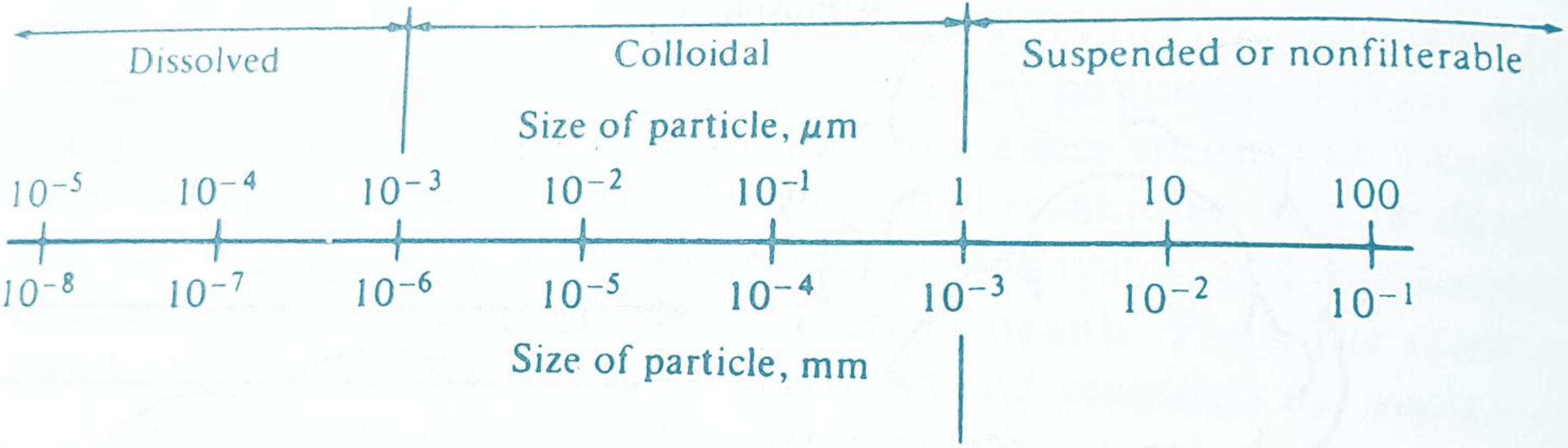
TVS = total volatile solids

TFS = total fixed solids

Unit of Total solid= mg/L

Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018). Image source: Datastream, https://datastream.org/en-ca/guidebook/total-suspended-solids-tss-and-total-dissolved-solids-tds

Total Solids



Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Numerical

A filterable residue analysis is run on a sample of water as follows. Prior to filtering, the crucible and filter pad are kept overnight in the drying oven, cooled, and the dry mass (tare mass) of the pair determined to be 54.352 g. Two hundred and fifty milliliters of the sample is drawn through a filter pad contained in the porous-bottom crucible. The crucible and filter pad are then placed in a drying oven at 104°C and dried until a constant mass of 54.389 g is reached. Determine the suspended solids concentration of the sample.

<u>Determine the mass of solids removed.</u>

Tare mass + solids = 54.389 g

- Tare mass = 54.352 g

Mass of solids = 0.037 g = 37 mg

<u>Determine the concentration of the solids.</u>

$$\text{Total Suspended Solids (mg/L)} = \frac{(Wf - Wi) \times 1000}{\text{Volume of sample, mL}}$$

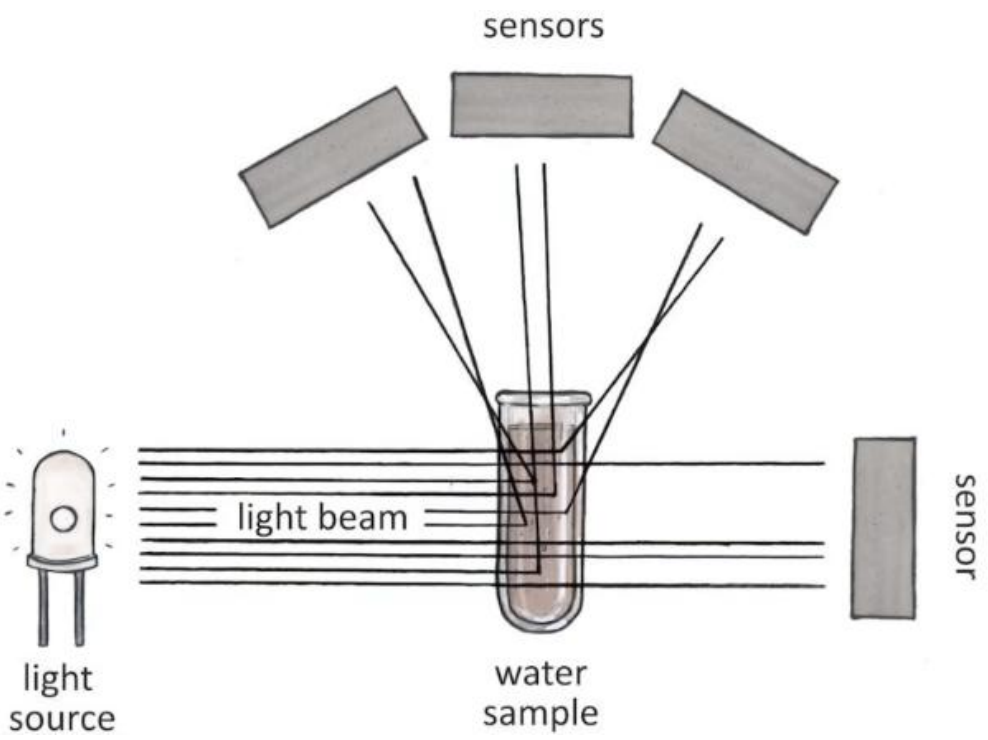
$$= \frac{37 \times 1000}{250}$$

$$= 148 \text{ mg L}$$

Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Turbidity

- A direct measurement of suspended solids is not usually performed on samples from natural bodies of water or on potable (drinkable) water supplies. The nature of the solids in these waters and the secondary effects they produce are more important than the actual quantity. For such waters a test for turbidity is commonly used.
- **Turbidity is a measure of the extent to which light is either absorbed or scattered by suspended material in water.** Because absorption and scattering are influenced by both size and surface characteristics of the suspended material, turbidity is not a direct quantitative measurement of suspended solids. For example, one small pebble in a glass of water would produce virtually no turbidity. If this pebble were crushed into thousands of particles of colloidal size, a measurable turbidity would result, even though the mass of solids had not changed.



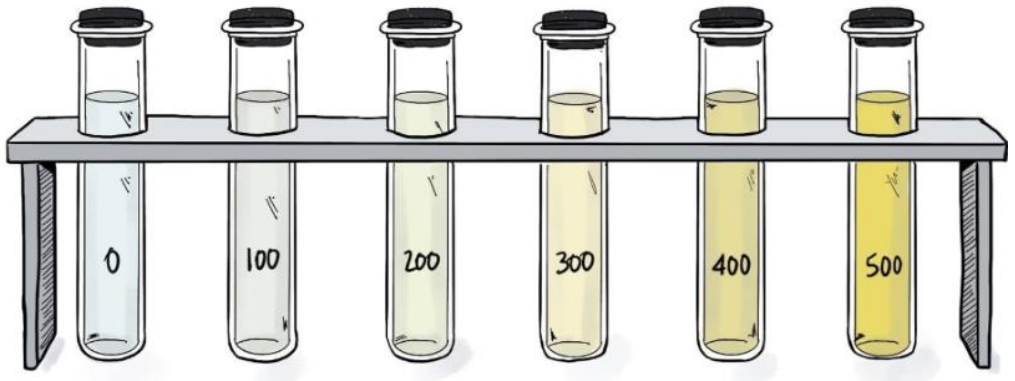
Turbidity is expressed in Nephelometric Turbidity Units (NTU).

Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018). Image source: Datastream, https://datastream.org/en-ca/guidebook/total-suspended-solids-tss-and-total-dissolved-solids-tds

Color

- Pure water is colorless, but water in nature is often colored by foreign substances Water whose color is partly due to suspended matter is said to have apparent color. Color contributed by dissolved solids that remain after removal of suspended matter is known as true color.

- Although several methods of color measurement are available, methods involving comparison with standardized colored materials are most often used. Color- comparison tubes containing a series of standards may be used for direct comparison of water samples that have been filtered to remove apparent color. Results are expressed in true color units (TCUs) where one unit is equivalent to the color produced by 1 mg/L of platinum in the form of chlorplatinate ions.



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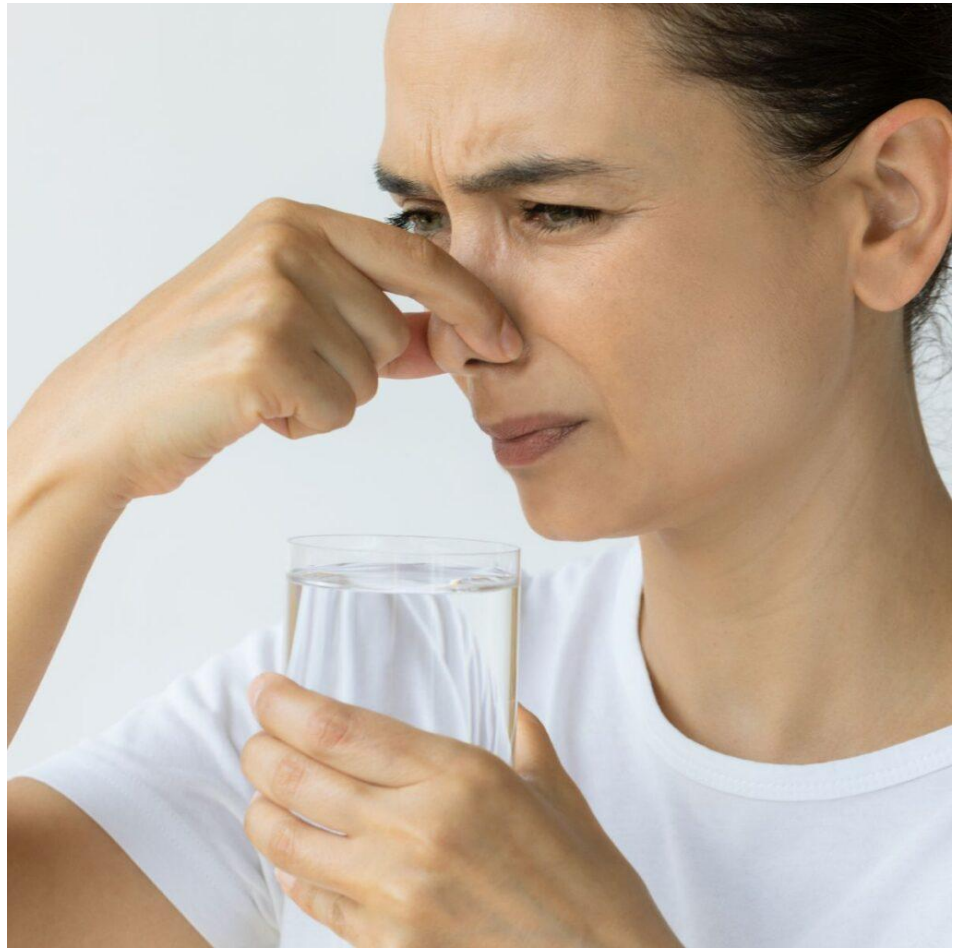
Taste and odor

- The terms taste and odor are themselves definitive of this parameter. Because the sensations of taste and smell are closely related and often confused, a wide variety of tastes and odors may be attributed to water by consumers. Substances that produce an odor in water will almost invariably impart a taste as well. The con- verse is not true, as there are many mineral substances that produce taste but no odor.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Aquazona, https://www.aquazona.com/common-water-problems/bad-water-taste-smell/, Accessed on: 14 January, 2025

Taste and odor

- Quantitative tests that employ the human senses of taste and smell can be used for this purpose. An example is the test for the threshold odor number (TON) Varying amounts of odorous water are poured into containers and diluted with enough odor-free distilled water to make a 200-mL mixture. An assembled panel of five to ten "noses" is used to determine the mixture in which the odor is just barely detectable to the sense of smell. The TON of that sample is then calculated, using the formula:

$$TON = \frac{A + B}{A}$$

where A is the volume of odorous water (mL) and B is the volume of odor-free water required to produce a 200-mL mixture.

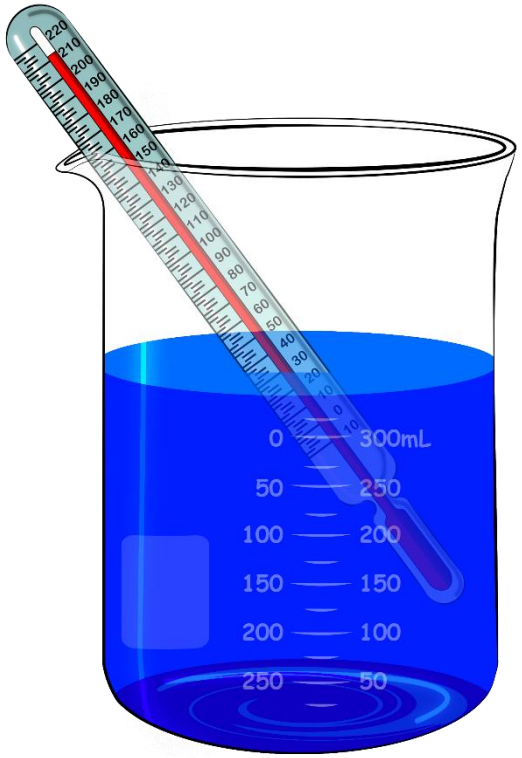
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Bing Search, CCBY

Temperature

- Temperature is not used to evaluate directly either potable water or wastewater. It is, however, one of the most important parameters in natural surface-water systems. The temperature of surface waters governs to a large extent the biological species present and their rates of activity. Temperature has an effect on most chemical reactions that occur in natural water systems. Temperature also has a pronounced effect on the solubilities of gases in water.

- An increase of 10°C is usually sufficient to double the biological activity if essential nutrients are present. At elevated temperatures and increased metabolic rates, organisms that are more efficient at food utilization and reproduction flourish, while other species decline and are perhaps eliminated altogether.

- Accelerated growth of algae often occurs in warm water and can become a problem when cells cluster into algae mats. Natural secretion of oils by algae in the mats and the decay products of dead algae cells can result in taste and odor problems.



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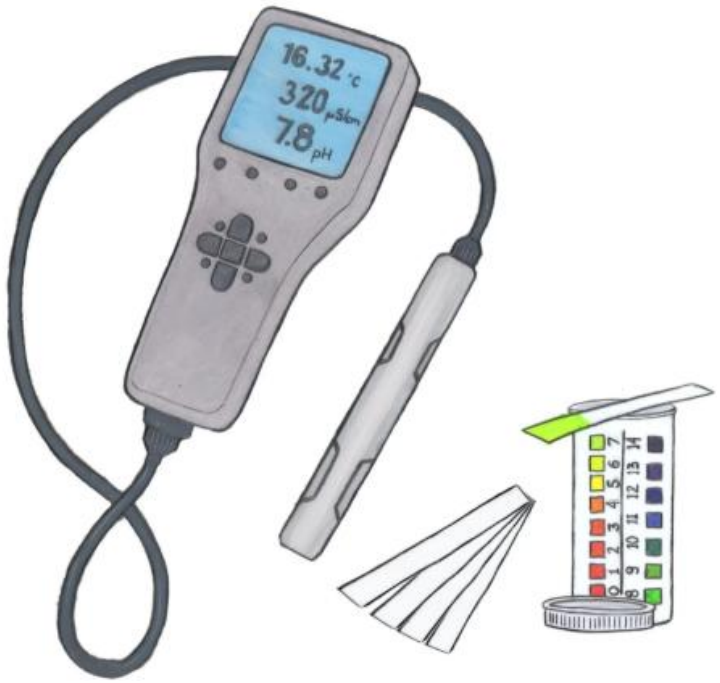
pH

- pH measures the degree to which water is acidic (like lemon juice) or basic (like bleach or soap). pH is measured on a scale that ranges from 0 (strongly acidic) to 14 (strongly basic). In the middle is 7, where the pH is neutral (like in pure water).

- It is the common logarithm of the reciprocal of the hydrogen ion concentration. Completely neutral water measures 7.0 on the scale. The values of pH lower than 7.0 indicate the hydrogen ion concentration is greater than the hydroxyl ion concentration, and the water is termed acidic. The opposite condition is implied when the pH exceeds 7.0, and water is termed basic.

- pH can be assessed using pH strips that change colour based on how acidic or basic the water is. The colour is evaluated against a guide to determine the approximate pH range of the water. The colour is evaluated against a guide to determine the approximate pH range of the water. pH can also be measured directly in a lake or river using a water quality meter with a pH sensor based on principle of Potentiometry. Before use, this sensor is calibrated using standard solutions of known pH (typically pH 4, 7 and 10).



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Datastream, https://datastream.org/en-ca/guidebook/ph , Accessed on: 14 January, 2025

Numerical

Question No. 1: An analysis for suspended solids is run as follows: (1) A fiberglass filter is dried to a constant mass of 0.137 g; (2) 100 mL of a sample is drawn through the filter; and (3) the filter and residue are placed in a drying over at 104°C until a constant mass of 0.183 g is reached. Determine the suspended-solids concentration in milligrams per liter.

Question No. 2: One hundred milliliters of the filtrate from a suspended-solids analysis is placed in an evaporation dish whose tare mass has been determined to be 327.485 g. The contents of the dish are evaporated to dryness, and the total mass of the dish and solids is found to be 327.517 g. Determine the quantity of filterable residue (in milligrams per liter).

Source: 1. Howard S. Peavy et al, Environmental Engineering, McGraw Hill Education (2018).

Thank you

For any questions, please feel free to reach out to me.